Capstone Project: Basic Web Vulnerability Scanner (Python)

Project Overview

In this project, you will develop a simple Python tool designed to identify basic security weaknesses in small websites. The tool will focus on analyzing web forms for common vulnerabilities such as SQL Injection and Cross-Site Scripting (XSS).

The objective is to understand how input-based attacks work and how basic security tools detect unsafe input handling in web applications.

Project Objective

Develop a Python script that is capable of:

- Accepting a website URL as input
- Identifying forms on the webpage
- Sending both normal and malicious test inputs
- Analyzing the website's response
- Reporting potential security risks

Core Requirements

1. Website Input

The program must:

- Accept a URL from the user
- Load and retrieve the webpage content
- Validate that the URL is accessible before scanning

2. Form Detection

The tool must be able to:

- Identify HTML forms within the webpage
- Extract form details, including:
    - Form action (submission endpoint)
    - Input fields (e.g., text, password, email)

Hint: Use BeautifulSoup for HTML parsing

3. Basic SQL Injection Testing

For each detected input field, the tool must:

- Send normal input values (e.g., test123)
- Send SQL injection payloads such as:
  - ' OR '1'='1
  - admin' --

The tool should analyze responses for:

- Database error messages
- Unexpected changes in response content

If any of these are detected, flag the form as a possible SQL Injection vulnerability.

4. Basic Cross-Site Scripting (XSS) Testing

The tool must test input fields using XSS payloads such as:

- <script>alert(1)</script>

The response should be checked for:

- Reflected script content in the output
- Lack of input sanitization

If detected, flag the input as a possible XSS vulnerability.

5. Reporting

At the end of the scan, the tool must generate a report that includes:

- Target URL
- Number of forms detected
- Inputs tested
- Identified vulnerabilities

The report must be displayed in the terminal. Optionally, results may be saved to a .txt file.

6. Ethical Constraint

The tool must:

- Only scan websites explicitly provided by the user
- Not performing automatic scanning of external or random websites
- Display a warning message stating:
  "This tool is for educational purposes only"

Submission Deadline

All submissions must be completed between **28th May, 2026 and 31st May 2026**.